Lab 12

Associating Roles with User Store Groups

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue.

Training Objective

Learn how to use groups in a user store (here, the WSO2.COM LDAP from the previous lab) as roles in WSO2 API Manager, and how to map them to the built-in roles that control portal access.

Business Scenario

PizzaShack wants to reuse existing user store groups to maintain access control on its API Management platform.

High-Level Steps

- Create LDAP groups as roles with permissions
- Map them to the Internal/* roles in the Admin Portal
- Check the access of each user

Before You Start

This lab needs the **WSO2.COM** LDAP user store from the previous lab. Because WSO2.COM is an LDAP, every role created in it is stored as an LDAP group, and an LDAP group cannot be empty. So first create one user per role, then create the roles with that user as a member.

Part 1 – Create the LDAP Users and Roles

1. Open the Management Console and log in as admin:

```
https://<APIM_HOST>:<HTTPS_PORT>/carbon
```

Example:

```
https://localhost:9443/carbon
```

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2. Go to **Main > Users and Roles > Add > Add New User** and create three users in the **WSO2.COM** domain without roles: `ldaparch`, `ldapdev` and `ldapadmin`.

3. Go to **Add > Add New Role**, select the domain **WSO2.COM**, enter the role name and click **Next**. Select the permissions in the table below and click **Next**. Select the user for that role and click **Finish**. Without a user, the role cannot be created (error: `Can not create empty role`).

4. Repeat step 3 for the three roles:

Role	Permissions	User
Architect	Admin Permissions -> Login Admin Permissions -> Manage -> Search Admin Permissions -> Manage -> Manage Tiers Admin Permissions -> Manage -> API Admin Permissions -> Manage -> Resources	`ldaparch`
Developer	Admin Permissions -> Login Admin Permissions -> Manage -> Search Admin Permissions -> Manage -> API -> Subscribe Admin Permissions -> Manage -> API -> Create Admin Permissions -> Manage -> Resources	`ldapdev`
Admin	All Permissions	`ldapadmin`

Part 2 – Map the LDAP Groups to the Built-in Roles

The Publisher and Developer Portal call REST APIs whose resources are protected by scopes. Each UI action (login, create API, publish, subscribe) needs a scope. These scopes are mapped to the built-in Internal/* roles (Internal/creator, Internal/publisher, and so on). So, to give the LDAP groups access to the portals, map each group to the matching built-in role.

5. Log in to the Admin Portal and go to **Settings > Scope Assignments**:

```
https://<APIM_HOST>:<HTTPS_PORT>/admin
```

Example:

```
https://localhost:9443/admin
```

6. Click **Add scope mappings**, enter the **Role Name** (for example `WSO2.COM/Architect`), click **Next**, select **Role alias**, pick the **Mapping role** from the table below and click **Save**. Repeat for each role. The Mapping role field accepts more than one role: for `WSO2.COM/Developer` select both **Internal/creator** and **Internal/subscriber**.

Role Name (LDAP group)	Mapping role (built-in)
`WSO2.COM/Architect`	`Internal/publisher`
`WSO2.COM/Developer`	`Internal/creator, Internal/subscriber`
`WSO2.COM/Admin`	`admin`

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Part 3 – Check the Result

7. Log in to the Publisher (and the Developer Portal) as each user:

```
https://<APIM_HOST>:<HTTPS_PORT>/publisher
https://<APIM_HOST>:<HTTPS_PORT>/devportal
```

Example:

```
https://localhost:9443/publisher
https://localhost:9443/devportal
```

- `WSO2.COM/ldaparch`: can publish APIs but not create them.
- `WSO2.COM/ldapdev`: can create APIs, and subscribe in the Developer Portal.
- `WSO2.COM/ldapadmin`: full access, including the Admin Portal.

Expected Outcome

Groups in the WSO2.COM LDAP are used as API Manager roles with their own permissions, and are mapped to the built-in Internal/* roles so that their members get the right access to the Publisher, Developer Portal and Admin Portal.

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